Mathematical Induction

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Mathematical Induction

INTRODUCTION

In algebra, there are certain results that are formulated with n number of terms, where n is a natural number (i.e. a positive integer). Those results can be proved by a specific technique, known as the **Principle of Mathematical Induction**. We use the symbol **P(n)** (read "P of n") to denote some proposition which depends on the positive integer n. For example, P(n) might denote the sum of the first n odd positive numbers, that is

$$1+3+5+7+9+...+(2n-1)=n^2$$

Where $n=1,2,3,...,n$.

FIRST PRINCIPLE OF MATHEMATICAL INDUCTION:

The proof of the proposition P(n) by mathematical induction for all $n \in N$ consists of the following three steps:

Step-1 : Verification step

Verify that the proposition P(n) is true for n = 1, i.e., the first natural or the smallest position integer. This is also called the basic step of the induction.

Step-2 : Induction step

Assume that the proposition will also be true for some $n = k \geq 1$, i.e. we assume P(k) to be true. This is called the induction step.

Step-3 : Generalization step

If P(k) is true, then prove that the proposition is also true for n=(k+1), which is the next position integer (i.e. the next natural number), i.e. we have to prove that P(k+1) must also be true. In this step we prove that the implication $P(k) \Rightarrow P(k+1)$ is true.

Next, we generalize the result be saying that, since the proposition is proved to be true for n = k+1, then it must also be true for n=k, and hence the proposition will be true for all n belonging to the set of natural numbers.

SECOND PRINCIPLE OF MATHEMATICAL INDUCTION:

Step-1 : Verification step

Assume that the proposition P(n) is true for n = r, where r is some fixed integer.

Step-2 : Induction step

Assume that the proposition P(n) is true for n = r, r+1, r+2, ..., m (where m > r, m ∈ I)

Step-3 : Generalization step

Prove that the proposition P(n) is true for n = m + 1. Thus, if true, we generalize the result by saying that since the proposition is true for n = m + 1, then it must also be true for n=r, r+1, r+2, ...,m as assumed in Step2. Thus, the proposition is true for all $n \geq r$ belonging to the set of natural numbers.

SOLVED EXAMPLE

Q. **By mathematical induction, $\frac{1}{1.2.3}+\frac{1}{2.3.4}+...+\frac{1}{n(n+1)(n+2)}$ is equal to**

(A) $\frac{n(n+1)}{4(n+2)(n+3)}$ **(B)** $\frac{n(n+3)}{4(n+1)(n+2)}$

(C) $\frac{n(n+2)}{4(n+1)(n+3)}$ **(D) None of these**

Sol: **(B)**

Let $P(n): \frac{1}{1.2.3}+\frac{1}{2.3.4}+...+\frac{1}{4(n+1)(n+2)}=\frac{n(n+3)}{4(n+1)(n+2)}$

(i) For n = 1

$L.H.S. = \frac{1}{1.2.3}=\frac{1}{6}$ and $R.H.S. = \frac{1(1+3)}{4(1+1)(1+2)}=\frac{1}{6}$

$\therefore$ P(1) is true.

(ii) Let P(k) be true, then

$P(k): \frac{1}{1.2.3}+\frac{1}{2.3.4}+...+\frac{1}{k(k+1)(k+2)}$

$=\frac{k(k+3)}{4(k+1)(k+2)}$ ...(1)

(iii) For n = k + 1,

$P(k+1): \frac{1}{1.2.3}+\frac{1}{2.3.4}+...+\frac{1}{k(k+1)(k+2)}+\frac{1}{(k+1)(k+2)(k+3)}=\frac{(k+1)(k+4)}{4(k+2)(k+3)}$

$$\text{L.H.S.} = \frac{1}{1.2.3} + \frac{1}{2.3.4} + \ldots + \frac{1}{k(k+1)(k+2)} + \frac{1}{(k+1)(k+2)(k+3)}$$

$$= \frac{k(k+3)}{4(k+1)(k+2)} + \frac{1}{(k+1)(k+2)(k+3)} \qquad \text{[From Eq. (1)]}$$

$$= \frac{k(k+3)^2 + 4}{4(k+1)(k+2)(k+3)} = \frac{k^3 + 6k^2 + 9k + 4}{4(k+1)(k+2)(k+3)} = \frac{(k+1)^2(k+4)}{4(k+1)(k+2)(k+3)}$$

$$= \frac{(k+1)(k+4)}{4(k+2)(k+3)} = \text{R.H.S.}$$

Hence, P(k+1) is true.
Hence, by the principle of mathematical induction for all $n \in N, P(n)$ is true.

Q. **For all $n \in N, (3)(5^{2n+1}) + 2^{3n+1}$ is divisible by**

(A) 17 **(B) 19** **(C) 21** **(D) 23**

Sol: **(A)**

$P(n) : 3(5^{2n+1}) + 2^{3n+1}$

$P(1) : 3(5^3) + 2^4 = 3(125) + 16$

$= 375 + 16 = 391 = 17(23)$

So, P(1) is divisible by 17.
Let $P(k) : 3(5^{2k+1}) + 2^{3k+1} = 17m$

$\therefore P(k+1) : 3(5^{2k+3}) + 2^{3k+4} = 17\lambda$

L.H.S. of $P(k+1) = 3(5^{2k+3}) + 2^{3k+4}$

$= 3(5^{2k+1})(5^2) + (2^{3k+1})(2^3)$

$= (17m - 2^{3k+1})(25) + 8(2^{3k+1})$ [Assuming P(k) to be true]

$= 17(25m) - 25(2^{3k+1}) + 8(2^{3k+1})$

$= 17(25m) - 17(2^{3k+1})$

$= 17(25m - 2^{3k+1})$

$$= 17\left(25m - 2^{3k+1}\right) = 17\lambda$$

Thus, P(k + 1) is divisible by 17 whenever P(k) is divisible by 17.
Hence P(n) is divisible by 17 for all $n \in N$.

Q. **If $x^{2n-1} + y^{2n-1}$ is divisible by x + y, then n is**

(A) a positive integer **(B) an even positive integer**
(C) an odd positive integer **(D) none of these**

Sol: **(A)**

$P(n) : x^{2n-1} + y^{2n-1}$

$P(1) : x^1 + y^1 = x + y$, which is divisible by $x + y$

Let $P(k) : x^{2k-1} + y^{2k-1} = (x + y)m$

$\therefore P(k+1) = x^{2k+1} + y^{2k+1} = (x + y)\lambda$

L.H.S. of $P(k+1) = x^{2k+1} + y^{2k+1}$

$$= x^2\left(x^{2k-1}\right) + y^2\left(y^{2k-1}\right)$$

$$= x^2\left[m(x+y) - y^{2k-1}\right] + y^2\left(y^{2k-1}\right) \quad \text{[Assuming P(k) to be true]}$$

$$= (x+y)\left(mx^2\right) - y^{2k-1}\left(x^2 - y^2\right)$$

$$= (x+y)\left[mx^2 - y^{2k-1}(x-y)\right] = (x+y)\lambda$$

Thus, P(k + 1) is divisible by (x + y) whenever P(k) is divisible by (x + y). Hence P(n) is divisible by x + y for all $n \in N$, i.e. for all positive integers.

Q. **When $P(n) = 9^n - 8^n$ is divided by 8, then the remainder is**

(A) 2 **(B) 3** **(C) 1** **(D) 7**

Sol: **(C)**

$P(n) = 9^n - 8^n$

$\therefore P(1) = 9 - 8 = 1$

$\therefore P(1) - 1 = 0$ which is divisible by 8

$\therefore$ 1 is the remainder when P(n) is divided by 8

Now, $P(2) = 9^2 - 8^2 = 17 = 16 + 1$

Remainder is 1, when divided by 8.

let remainder = 1 for $n \in k$ where k is positive integer

$\Rightarrow P(k) = 9^k - 8^k = 8m + 1$ (m is positive integer) ...(i)

Now let remainder = 1 for $P(k + 1) = 9^{k+1} - 8^{k+1}$

$\Rightarrow 9 \times 9^k - 8 \times 8^k$

$\Rightarrow 9(8^k + 8m + 1) - 8 \cdot 8^k$ from (i)

$\Rightarrow 9 \cdot 8^k - 8 \cdot 8^k + 9(8m + 1)$

$\Rightarrow 8^k + 8(8m + 1) + 8m + 1$

$\Rightarrow 8(8^{k-1} + (8m + 1) + m) + 1$

$\Rightarrow 8(8^k + 9m + 1) + 1$

$\Rightarrow$ Remainder is always 1 for $n \in Z^+$.

Q. **Let $P(n) = 2^{3n} - 7n - 1$, then P(n) is divisible by**

(A) 63 **(B) 36** **(C) 49** **(D) 25**

Sol. **(C)**

$P(n) : 2^{3n} - 7n - 1$

$P(1) : 2^3 - 7 - 1 = 8 - 7 - 1 = 0$

$P(2) : 2^6 - 7(2) - 1 = 64 - 14 - 1 = 49$

Which is divisible by 49

Let $P(k) : 2^{3k} - 7k - 1 = 49m$

$\therefore P(k + 1) = 2^{3k+3} - 7(k + 1) - 1 = 49\lambda$

L.H.S. of $P(k + 1) = 2^{3k+3} - 7(k + 1) - 1$

$= (2^{3k})(2^3) - 7k - 7 - 1$

$= (49m + 7k + 1)(8) - 7k - 8$ [Assuming P(k) to be true]

$= 49(8m) + 56k + 8 - 7k - 8$

$= 49(8m + k) = 49\lambda$

Thus, P(k + 1) is divisible by 49 whenever P(k) is divisible by 49.

Hence P(n) is divisible by 49 for all $n \in N$.

Q. **Let P(n): $n^2 - n + 41$ is a prime number, then**

(A) P(1) is not true **(B)P(5) is not true**

(C) P(9) is not true **(D) P(41) is not true**

Sol. **(D)**

$P(n) = n^2 - n + 41$

for $P(1) = 1 - 1 + 41 = 41$ which is a prime Number

$P(5) = 25 - 5 + 41 = 66 - 5 = 61$ which is a Prime Number

$P(9) = 81 - 9 + 41 = 117$ which is a prime number

$P(41) = (41)^2 - 41 + 41 = 1681$ which is Not a prime numbers

$\Rightarrow$ P(41) is Not true

Q. **Let P(n): $n^2 + n$ is odd, then $P(n) \Rightarrow P(n + 1)$ for all n. and P(1) is not true. From here, we can conclude that**

(A) P(n) is true for all $n \in N$ **(B) P(n) is true for all $n \geq 2$**

(C) P(n) is false for all $n \in N$ **(D) P(n) is true for all $n \geq 3$**

Sol. **(C)**

$P(n)$: $n^2 + n$ is odd

then $P(n) = P(n + 1)$ for all n

Here P(1) = 2 which is not odd

Hence P(1) is Not true

for n = 2, $P(2) = 4 + 2 = 6$ which is not odd

for n = 3, $P(2) = 9 + 3 = 12$ which is not odd

Hence options (A), (B), (D) are Not true

Q. **Consider the statement P(n): $n^2 \geq 100$. Here. $P(n) \Rightarrow P(n + 1)$ for all n. Does it mean that**

(A) P(n) is true for all n **(B) P(n) is true for all $n \geq 2$**

(C) P(n) is true for all $n \geq 3$ **(D) None of these**

Sol. **(D)**

$P(n)$: $n^2 \geq 100$

$P(n) \Rightarrow P(n + 1)$ for all n

For $n^2 \geq 100 \Rightarrow n \geq 10$ or $n \leq -10$

Q. **$9^n - 8n - 1$ is divisible by 64 is**
(A) Always true
(B) Always false
(C) Always true for rational values of n
(D) Always true irrational value of n

Sol. **(A)**
$9^n - 8n - 1$
For n = 1, 9.9 = 0 which is divisible by 64
For n = 2, 81 – 196 – 1 = 81 – 17 = 64 which is divisible by 64
$\Rightarrow 9^n - 8n - 1$ is true for all n

Q. **$2^n > 2n + 1$ for all natural numbers n is**
(A) True **(B) False**
(C) Cannot be determined **(D) Depends on n**

Sol. **(B)**
$2^n > 2n + 1$ for all Natural Number n
for n = 1, 2 > 3 which is Not true
for n = 2, 4 > 5 which is Not true
Hence $2^n > 2n + 1$ is not true

Q. **For all natural number n, $3^{2n} - 1$ is divisible by**
(A) 3 **(B) 5** **(C)6** **(D) 8**

Sol. **(D)**
For natural number n, $3^{2n} - 1$
For n = 1, $3^2 - 1 = 8$ which is divisible by 8
For n = 2, $3^4 - 1 = 81 - 1 = 80$ which is divisible by 8
$\Rightarrow 3^{2n} - 1$ is divisible by 8

Q. **The statement $2^n > n^2$ (where $n \in N$) is true for**
(A) All n **(B) $n \geq 3$** **(C) $n < 4$** **(D) $n > 4$**

Sol. **(D)**
$2^n > n^2$
For n = 1, 2 > 1 which is true
For n = 2, 4 > 4 which is Not true
For n = 3, 8 > 9 which is Not true
For n = 4, 16 > 16 which is Not true

For n = 5, 32 > 25 which is true
$\Rightarrow 2^r > n^2$ for n > 4

Q. **Choose the statement which is correct for all n ∈ N**

(A) $2^n > 2n + 1$ **(B)** $x^n - y^n$ **is divisible by** $(x + y)$

(C) $x^n - y^n$ **is divisible by** $(x-y)$ **(D)** $1+\frac{1}{4}+\frac{1}{9}+\frac{1}{16}+\ldots+\frac{1}{n^2}<2-\frac{1}{n}$

Sol. **(C)**

$x^n - y^n = (x-y)\left(x^{n-1} + x^{n-2}y + \ldots + y^{n-1}\right)$

$\Rightarrow x^r - y^n$ is divisible by x−y

Q. **For each n ∈ N,** $3^{2n} - 1$ **is divisible by**

(A) 8 **(B)16** **(C) 32** **(D) 10**

Sol. **(A)**

$3^{2n} - 1$

For n = 1, $3^2 - 1 = 8$ which is divisible by 8

For n = 2, $3^4 - 1 = 80$ which is divisible by 8

$\Rightarrow 3^{2n} - 1$ is divisible by 8

Q. **The statement** $n! > 2^{n-1}$, $n \in N$ **is true for**

(A) $n > 1$ **(B)** $n > 2$ **(C) All n** **(D) No n**

Sol. **(B)**

$n! > 2^{n-1}$

For n = 1, $1! > 2^{1-1} \Rightarrow 1 > 1$

Which is Not true

For n = 2

$2! > 2^1$ which is Not true

For n = 3, $3! > 2^{3-1}$

$6 > 4$ which is true

$n! > 2^{n-1}$, n ∈ N is true for n > 2

Q. **For each n ∈ N,** $3 \cdot (5^{2n+1}) + 2^{3n+1}$ **is divisible by**

(A) 17 **(B) 19** **(C) 21** **(D) 23**

Sol. **(A)**

$3 \cdot (5^{2n+1}) + 2^{3n+1}$, $n \in N$

For $n = 1$, $3.5^3 + 2^4$

$375 + 16 = 391$

Which is divisible by 17

Q. **The proposition among the following that is not true for all $n \in N$, is**

(A) $2^n > n$ **(B)** $2^n > n^2$

(C) $1 + 2 + 3 + \ldots + n < (2n + 1)^2$ **(D)** $(2n + 7) < (n + 3)^2$

Sol. **(B)**

$2^n > n$

For $n = 1$, $2 > 1$ which is true

For $n = 2$, $2^2 > 2 \Rightarrow 4 > 2$ which is true

$\Rightarrow 2^n > n$ is true for all n

Now, $2^n > n^2$

For $n = 1$, $2 > 1$ which is true

For $n = 2$, $4 > 4$ which is Not true

$\Rightarrow 2^n > n^2$ is Not always true

Q. **The proposition that is not true for $n > 1$ ($n \in N$), is**

(A) $1+\frac{1}{4}+\frac{1}{9}+\ldots+\frac{1}{n^2}<2-\frac{1}{n}$ **(B)** $1+\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{3}}+\ldots+\frac{1}{\sqrt{n}}>\sqrt{n}$

(C) $\frac{1}{n+1}+\frac{1}{n+2}+\ldots+\frac{1}{2n}>\frac{13}{24}$ **(D)** $\frac{1}{2}\cdot\frac{3}{4}\cdot\frac{5}{6}\ldots\frac{2n-1}{2n}>\frac{1}{\sqrt{3n+1}}$

Sol. **(D)**

$$1+\frac{1}{4}+\frac{1}{9}+\ldots+\frac{1}{n^2}<2-\frac{1}{n}$$

For $n = 2$, $1+\frac{1}{4}<2-\frac{1}{2}$

$$\frac{5}{4}<\frac{3}{2}$$

$1.25 < 1.5$ which is true

$\Rightarrow 1+\frac{1}{4}+\frac{1}{9}+\ldots+\frac{1}{n^2}<2-\frac{1}{n^2}$ is true for $n > 1$

Now, $1+\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{3}}+\ldots+\frac{1}{\sqrt{n}}>\sqrt{n}$

For n = 2

$1+\frac{1}{\sqrt{2}}>\sqrt{2}$

Which is true for n > 1

Now, $\frac{1}{n+1}+\frac{1}{n+2}+\ldots+\frac{1}{2n}>\frac{13}{24}$

For n = 2 $\quad \frac{1}{2}+\frac{1}{4}>\frac{13}{24}$

$\frac{3}{4}>\frac{13}{24}$

$\Rightarrow$ which is true for n > 1

Now, $\frac{1}{2}.\frac{3}{4}.\frac{5}{6}....\frac{2n-1}{2n}>\frac{1}{\sqrt{3n+1}}$

For n = 2

$\frac{1}{2}.\frac{3}{4}>\frac{1}{2}$

Which is Not true for n > 1

Q. **Which of the following is true for n $\in$ N?**

(A) $\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\ldots+\frac{1}{2^n}<1$

(D) $\frac{1}{n+1}+\frac{1}{n+2}+\ldots+\frac{1}{3n+1}<1$

(C) $n^4>10^n$

(D) $^{2n}C_n>\frac{4^n}{n+1}$

Sol. **(A)**

$\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\ldots+\frac{1}{2n}<1$

For $n=1, \frac{1}{2}<1$ which is true

For $n=2, \frac{1}{4}<1$ which is true

$\Rightarrow \frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\ldots+\frac{1}{2n}<1$ is true for all n $\in$ N

Q. **Prove that $7^{2n} + (2^{3n-3})(3^{n-1})$ is divisible by 25 for any natural number n.** **[JEE Adv. 1982]**

Sol. P(n) : $7^{2n} + 2^{3n-3} \cdot 3^{n-1}$ is divisible by 25 $\forall$ n $\in$ N

Let us prove it by Mathematical Induction .

P(1) : $7^2 + 2^0 . 3^0 = 49 + 1 = 50$ which is divisible by 25.

$\therefore$ P(1) is ture

Let P(k) is true that $7^{2k} + 2^{3k-3} \cdot 3^{k-1}$ is divisible by 25

$\Rightarrow 7^{2k} + 2^{3k-3} . 3^{k-1} = 25m$ where m $\in$ Z.

$\forall\ 2^{3k-3} . 3^{k-1} = 25m - 7^{2k}$ ... (1)

Consider P(k + 1)

$7^{2(k+1)} + 2^{3(k+1)-3} . 3^{k+1-1} = 7^{2k} . 7^2 + 2^{3k} . 3^k$

$= 49.7^{2k} + 2^3 . 3 . 2^{3k-3} . 3^{k-1}$

$= 49.7^{2k} + 24(25m - 7^{2k})$

(Using eq. (1))

$= 49.7^{2k} + 24 \times 25m - 24 \times 7^{2k}$

$= 25.7^{2k} + 24 \times 25m = 25(7^{2k} + 24m)$

= 25 × some integral value which is divisible by 25.

$\therefore$ P(k + 1) is also true.

Q. **Use mathematical Induction to prove : If n is any odd positive integer, then $n(n^2 - 1)$ is divisible by 24.** **[JEE Adv. 1983]**

Sol. P(n) : $n(n^2 - 1)$ is divisible by 24 for any odd + ve integer.

For n = 2m − 1, it can be restated as

P(m) : $(2m - 1)(4m^2 - 4m) = 4m(m - 1)(2m - 1)$

is divisible by 24 $\forall$ m $\in$ N

$\Rightarrow$ P(m) : m(m − 1)(2m − 1) is divisible by 6 $\forall$ m $\in$ N.

Here P(1) = 0, divisible by 6.

$\therefore$ P(1) is true

Let it be true for m = k, i.e.,

$k(k - 1)(2k - 1) = 6p$

$\Rightarrow 2k^3 - 3k^2 + k = 6p$ ...(1)

consider P(k + 1) : $K(k + 1)(2k + 1) = 2k^3 + 3k^2 + k$

$= 6p + 3k^2 + 3k^2$ (Using(1))

$= 6(p + k^2) \Rightarrow$ divisible by 6

$\Rightarrow P(k + 1)$ is also true.

Q. **If p be a natural number then prove that $p^{n+1} + (p + 1)^{2n-1}$ is divisible by $p^2 + p + 1$ for every positive integer n.** **[JEE Adv. 1984]**

Sol. $P(n) : P^{n+1} + (p + 1)^{2r-1}$ is divisible by $p^2 + p + 1$

For $n = 1$, $P(1) : p^2 + p + 1$ which is divisible by $p^2 + p + 1$

$\therefore P(1)$ is true

Let $P(k)$ be true, i.e.,

$p^{k+1} + (p + 1)^{2k-1}$ is divisible by $p^2 + p + 1$

$\Rightarrow p^{k+1} + (p + 1)^{2k-1} = (p^2 + p + 1)m$ ... (1)

Consider $P(k + 1) : p^{k+2} + (p + 1)^{2k+1} = p \cdot p^{k+1} + (p + 1)^{2k-1} \cdot (p + 1)^2$

$= p[m(p^2 + p + 1) - (p + 1)^{2k-1}] + (p + 1)^{2k-1}(p + 1)^2$

$= p(p^2 + p + 1)m - (p + 1)^{2k-1} p + (p + 1)^{2k-1}(p^2 + 2p + 1)$

$= p(p^2 + p + 1)m + (p + 1)^{2k-1}(p^2 + p + 1)$

$= (p^2 + p + 1)[mp + (p + 1)^{2k-1}]$

$= (p^2 + p + 1)$ some integral value

$\therefore$ divisible by $p^2 + p + 1$ $\therefore P(k + 1)$ is also true

Hence by principle of mathematical induction $P(n)$ is true.

$\forall\, n \in N$

Q. **Use method of mathematical induction $2.\ 7^n + 3.\ 5^n - 5$ is divisible by 24 for all $n > 0$** **[JEE Adv. 1985]**

Sol. Let $A_n = 2.7^n + 3.5^n - 5$

Then $A_1 = 2.7 + 3.5 - 5 = 14 + 15 - 5 = 24$

Hence A_1 is divisible by 24.

Now assume that A_m is divisible by 24 so that we may write

$A_m = 2.7^m + 3.5^m - 5 = 24k, k \in N$ ...(1)

Then $A_{m+1} - A_m = 2\left(7^{m+1} - 7^m\right) + 3\left(5^{m+1} - 5^m\right) - 5 + 5$

$= 2.7^m(7 - 1) + 3.5^m(5 - 1) = 12 \cdot \left(7^m + 5^m\right)$

Since 7^m and 5^m are odd integers $\forall m \in N$, their sum must be an even integer, say $7^m + 5^m = 2p, p \in N$

Hence $A_{m+1} - A_m = 12.2p = 24p$

or $A_{m+1} = A_m + 24p = 24k + 24p$ [by (1)]

Hence A_{m+1} is divisible by 24

It follows by mathematical induction that A_n is divisible by 24 for all $n \in N$.

Q. **Prove by mathematical induction that $\frac{(2n)!}{2^{2n}(n!)^2} \le \frac{1}{(3n+1)^{1/2}}$ for all positive integers n.** **[JEE Adv. 1987]**

Sol. Let $P(n): \frac{(2n)!}{2^{2n}(n!)^2} \le \frac{1}{(3n+1)^{1/2}}$

For $n = 1, P(1): \frac{2!}{2^2(1!)^2} \le \frac{1}{(3+1)^{1/2}}$

$\Rightarrow \frac{1}{2} \le \frac{1}{2}$ which is true for n = 1

Assume that p(k) is true, then

$P(k): \frac{(2k)!}{2^{2k}(k!)^2} \le \frac{1}{(3k+1)^{1/2}}$

For $n = k + 1$,

$$\frac{[2(k+1)]!}{2^{2(k+1)}[(k+1)!]^2} = \frac{(2k+2)!}{2^{2k+2}[(k+1)!]^2}$$

$$= \frac{(2k+2)(2k+1)(2k)!}{4.2^{2k}(k+1)^2(k!)^2} \le \frac{(2k+2)(2k+1)}{4(k+1)^2} \cdot \frac{1}{(3k+1)^{1/2}}$$

[Using Induction hypothesis (1)]

$$= \frac{(2k+1)}{2.(k+1)(3k+1)^{1/2}}$$

Thus, $\frac{[2(k+1)]!}{2^{2(k+1)}[(k+1)!]^2} \le \frac{(2k+1)}{2(k+1)(3k+1)^{1/2}}$ ... (2)

In order to prove P(k + 1), it is sufficient to prove that

$\frac{(2k+1)}{2(k+1)(3k+1)^{1/2}} \le \frac{1}{(3k+4)^{1/2}}$ ... (3)

Squaring eq. (3), we get

$$\frac{(2k+1)^2}{4(k+1)^2(3k+1)} \le \frac{1}{3k+4}$$

$\Rightarrow (2k+1)^2(3k+4) - 4(k+1)^2(3k+1) \le 0$

$\Rightarrow (4k^2+4k+1)(3k+4) - 4(k^2+2k+1)(3k+1) \le 0$

$\Rightarrow (12k^3+28k^2+19k+4) - (12k^3+28k^2+20k+4) \le 0$

$\Rightarrow -k \le 0$

Which is true.

Hence from (2) and (3), we get

$$\frac{(2k+2)!}{2^{2k+2}[(k+1)!]^2} \le \frac{1}{(3k+4)1/2}$$

Hence the above inequation is true for $n = k+1$ and by the principle of induction it is true for all $n \in N$.

Q. **Let $p \ge 3$ be an integer and α, β be the roots of $x^2 - (p+1)x + 1 = 0$ using mathematical induction show that $\alpha^n + \beta^n$.** **[JEE Adv. 1992]**
(i) Is an integer **(ii) is not divisible by p**

Sol. Since α, β are the roots of $x^2 - (p+1)x = 1 = 0$
$\therefore \alpha + \beta = p + 1 ; \alpha\beta = 1$

Here $p \ge 3$ and $p \in Z$

(i) To prove that $\alpha^n + \beta^n$ is an integer.

Let us consider the statement "$\alpha^n + \beta^n$ is an integer,"

Then for $n - 1$, $\alpha + \beta = p + 1$ which is an integer, p being an integer.

$\therefore$ statement is true for $n = 1$.

Let us the statement be true for $n \le k$, i.e., $\alpha^k + \beta^k$ is an integer then,

$$\alpha^{k+1} + \beta^{k+1} = \alpha^k.\alpha + \beta^k \cdot \beta$$

$$= \alpha\left(\alpha^k + \beta^k\right) + \beta\left(\alpha^k + \beta^k\right) - \alpha\beta^k - \alpha^k\beta$$

$$= (\alpha+\beta)\left(\alpha^k + \beta^k\right) - \alpha\beta\left(\alpha^{k-1} + \beta^{k-1}\right.$$

$$= (\alpha+\beta)\left(\alpha^k + \beta^k\right) - \left(\alpha^{k-1} + \beta^{k-1}\right) \quad \text{... (1)}$$

[as $\alpha\beta = 1$]

= difference of two integers = some integral value

$\Rightarrow$ Statement is true for $n = k + 1$

$\therefore$ By the principle of mathematical induction the given statement is true for $\forall$ $n \in N$.

(ii) Let R_n be the remainder of $\alpha^n + \beta^n$ when divided by p where $0 \le R_n \le p - 1$

Since $\alpha + \beta = p + 1 \therefore R_1 = 1$

Also $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = (p + 1)^2 - 2$

$= p^2 + 2p - 1 = p(p + 1) + p - 1$

$\therefore R_2 = p - 1$

Also from equation (1) of previous part (i), we have

$$\alpha^{n+1} + \beta^{n+1} = (p+1)\left(\alpha^n + \beta^n\right) - \left(\alpha^{n-1} + \beta^{n-1}\right)$$

$$= p\left(\alpha^n + \beta^n\right) + \left(\alpha^n + \beta^n\right) - \left(\alpha^{n-1} + \beta^{n-1}\right)$$

$\Rightarrow R_{n+1}$ is the remainder of $R_n - R_{n-1}$ when divided by p

$\therefore$ We observe that $R_2 - R_1 = p - 1 - 1$

$\therefore R_3 = p - 2$

Similarly, R_4 is the remainder when $R_3 - R_2$ is divided by p where

$R_3 - R_2 = p - 2 - p + 1 = -1 = -p + (p - 1) \therefore R_4 = p - 1$

$R_4 - R_3 = p - 1 - p + 1 = 0 \quad \therefore R_5 = 1$

$R_5 - R_4 = 1 - p + 1 = -p + 2 \quad \therefore R_6 = p - 2$

It is evident for above that the remainder is either 1 or p – 1 or p – 2.

Since $p \ge 3$, so none is divisible by p.

Q. **Using mathematical induction, prove that**
$\tan^{-1}(1/3) + \tan^{-1}(1/7) + ... \tan^{-1}(1/(n^2 + n + 1)) = \tan^{-1}(n/(n + 2))$ **[JEE Adv. 1993]**

Sol. To prove $P(n) : \tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{7}\right) + ... \tan^{-1}\left(\frac{1}{n^2+n+1}\right) = \tan^{-1}\left(\frac{n}{n+2}\right)$

For n = 1, LHS $= \tan^{-1}\frac{1}{3}$

RHS $= \tan^{-1}\frac{1}{3} \Rightarrow$ LHS = RHS

$\therefore$ P(1) is true

Let P(k) be true, i.e.

$$\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{7}\right) + ... \tan^{-1}\left(\frac{1}{k^2+k+1}\right) = \tan^{-1}\left(\frac{k}{k+2}\right) \quad ...(1)$$

Consider P(k + 1)

$$\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{7} + ... \tan^{-1}\left(\frac{1}{k^2+k+1}\right) + \tan^{-1}\left(\frac{1}{(k+1)^2+(k+1)+1}\right)$$

$$= \tan^{-1}\left[\frac{k+1}{(k+1)+2}\right]$$

$$\text{LHS} = \tan^{-1}\left[\frac{k}{(k+2}\right] + \tan^{-1}\left(\frac{1}{\left(k^2+3k+3\right.}\right)$$

Using equation (1)

$$= \tan^{-1}\left[\frac{\frac{k}{(k+2} + \frac{1}{k^2+3k+3}}{1-\left(\frac{k}{k+2}\right)\left(\frac{1}{k^2+3k+3}\right)}\right]$$

$$= \tan^{-1}\left[\frac{(k+1)\left(k^2+2k+2\right)}{(k+3)\left(k^2+2k+2\right)}\right] = \tan^{-1}\left(\frac{k+1}{k+3}\right) = \text{RHS}$$

$\therefore$ P(k + 1) is also true.
Hence by the principle of mathematical induction
P(n) is true for every natural number.

Q. **If x is not an integral multiple of 2π use mathematical induction to prove that :**

$\cos x + \cos 2x + ... + \cos nx = \cos\frac{n+1}{2}x \sin\frac{nx}{2}\operatorname{cosec}\frac{x}{2}$ **[JEE Adv. 1994]**

Sol. Let P(n) : $\cos x + \cos 2x + ... + \cos nx$

$$= \cos\frac{n+1}{2}x \sin\frac{nx}{2}\operatorname{cosec}\frac{x}{2}$$

Where x is not an integral multiple of 2π
For n = 1 P(1) : L.H.S. = cosx

$$\text{R. H.S.} = \cos\frac{1+1}{2}x \sin\frac{x}{2}\operatorname{cosec}\frac{x}{2} = \cos x \qquad \ldots (1)$$

L.H.S. = R.H.S.
$\Rightarrow$ P(1) is true.
Let P(k) be true i.e.
$\cos x + \cos 2x + ... + \cos kx$

$$= \cos\frac{k+1}{2}x \sin\frac{kx}{2}\operatorname{cosec}\frac{x}{2} \qquad \ldots (2)$$

Consider P(k + 1)
$\cos x + \cos 2x + ... + \cos kx + \cos(k+1)x$

$$= \cos\left(\frac{k+2}{2}\right)x \sin\frac{(k+1)x}{2}\operatorname{cosec}\frac{x}{2}$$

L. H.S. [cosx + cos2x + ... + coskx + cos (k + 1)x

$$= \cos\left(\frac{k+1}{2}\right)x \sin\frac{kx}{2}\operatorname{cosec}\frac{x}{2} + \cos(k+1)x$$

[Using (2)]

$$= \left[\cos\left(\frac{k+1}{2}\right)x \sin\frac{kx}{2} + \cos(k+1)x \sin\frac{x}{2}\right]\operatorname{cosec}\frac{x}{2}$$

$$= \frac{1}{2}\left[2\cos\frac{(k+1)x}{2}\sin\frac{kx}{2} + 2\cos(k+1)x \sin\frac{x}{2}\right]\operatorname{cosec}\frac{x}{2}$$

$$= \frac{1}{2}\left[\sin\left(\frac{2k+1}{2}\right)x - \sin\frac{x}{2} + \sin\left(xk + \frac{3x}{2}\right) - \sin\left(xk + \frac{x}{2}\right)\right]\operatorname{cosec}\frac{x}{2}$$

$$= \frac{1}{2}\left[\sin\left(xk + \frac{3x}{2}\right) - \sin\frac{x}{2}\right]\operatorname{cosec}\frac{x}{2}$$

$$= \frac{1}{2}\left[2\cos\frac{(k+2)x}{2}\sin\frac{(k+1)x}{2}\right]\operatorname{cosec}\frac{x}{2}$$

$$= \cos\frac{(k+2)x}{2}\sin\frac{(k+1)x}{2}\operatorname{cosec}\frac{x}{2} = \text{R.H.S.}$$

∴ P(k + 1) is also true
Hence by the principle of mathematical induction
P(n) is true ∀ n ∈ N

Q. **Using mathematical induction prove that for every integer n ≥ 1, $(3^{2n} - 1)$ is divisible by 2^{n+2} but not by 2^{n+3}** **[JEE Adv. 1996]**

Sol. For n = 1, $3^{2^n} - 1 = 3^{2^1} - 1 = 9 - 1 = 8$
which is divisible by $2^{n+2} = 2^3 = 8$
but is not divisible by $2^{n+3} = 2^4 = 16$
Therefore, the result is true for n = 1.

Assume that the result is true for n = k. That is, assume that $3^{2^k} - 1$ is divisible by 2^{k+2} but is not divisible by 2^{k+3}

Since $3^{2^k} - 1$ is divisible by 2^{k+2} but not by 2^{k+3}, we can write $3^{2^k} - 1 = (m)2^{k+2}$ where m must be an odd positive integer, for otherwise $3^{2k} - 1$ will become divisible by 2^{k+3}

For n = K + 1 , we have $3^{2^{k+1}} - 1 = 3^{2^k \cdot 2} - 1 = \left(3^{2^k}\right)^2 - 1$ [Using(1)]

$= (m.\ 2^{k+2} + 1)^2 - 1$
$= m^2.(2^{k+2})^2 + 2m.\ 2^{k+2} + 1 - 1$
$= m^2.\ 2^{2k+4} + m.\ 2^{k+3} = 2^{k+3}(m^2.\ 2^{k+1} + m.)$

$\Rightarrow 3^{2^{k+1}} - 1$ is divisible by 2^{k+3}

But $3^{2^{k+1}} - 1$ is not divisible by 2^{k+4} for otherwise we must have 2 divides m^2. $2^{k+1} + m$.

But this is not possible as m is odd. thus, the result is true for $n = k + 1$

Q. **For every positive integer n, prove that $\sqrt{(4n+1)} < \sqrt{n} + \sqrt{n+1} < \sqrt{4n+2}$. Hence or otherwise, prove that $[\sqrt{n} + \sqrt{(n+1)}] = [\sqrt{4n+1}]$, where [x] denotes the greatest integer not exceeding x.** **[JEE Adv. 2000]**

Sol. For $n > 0, \sqrt{4n+1} > 0, \sqrt{n} + \sqrt{n+1} > 0$ and $\sqrt{4n+2} > 0$

Now, $\sqrt{4n+1} < \sqrt{n} + \sqrt{n+1} < \sqrt{4n+2}$ to be proved.

I. To prove $\sqrt{4n+1} < \sqrt{n} + \sqrt{n+1}$

Squaring both sides in $\sqrt{4n+1} < \sqrt{n} + \sqrt{n+1}$

$\Rightarrow 4n + 1 < n + n + 1 + 2\sqrt{n(n+1)}$

$\Rightarrow \quad 2n < 2\sqrt{n(n+1)} \Rightarrow n < \sqrt{n(n+1)}$ which is true.

II. To prove $\sqrt{n} + \sqrt{n+1} < \sqrt{4n+2}$

Squaring both sides,

$n + n + 1 + 2\sqrt{n(n+1)} < 4n + 2$

$\Rightarrow \quad 2\sqrt{n(n+1)} < 2n + 1$ Squaring again

$4[n(n+1)] < 4n^2 + 1 + 4n$ or $0 < 1$ which is true

Hence $\sqrt{4n+1} < \sqrt{n} + \sqrt{n+1} < \sqrt{4n+2}$

Further to prove $[\sqrt{n} + \sqrt{n+1}] = [\sqrt{4n+1}]$, we have to prove that there is no positive integer which lies between $\sqrt{4n+1}$ and $\sqrt{4n+2}$ or $[\sqrt{4n+1}] = [\sqrt{4n+2}]$. Using Mathematical induction.

We have to check $[\sqrt{4n+1}] = [\sqrt{4n+2}]$ for $n = 1$

$[\sqrt{5}] = [\sqrt{6}] \Rightarrow 2 = 2$, which is true

Assume for $n = k$ (arbitrary)

i.e., $[\sqrt{4k+1}] = [\sqrt{4k+2}]$

To prove for $n = k + 1$

To check $[\sqrt{4k+5}] = [\sqrt{4k+6}$ since $k \geq 0$

Here $4k + 5$ is an odd number and $4k + 6$ is even number. Their greatest integer will be different if $4k + 6$ is a perfect square that is $4k + 6 = r^2$

$\Rightarrow \quad k = \frac{r^2}{4} - \frac{6}{4}, \frac{6}{4}$ is not integer. But k has to be integer.

So 4k + 6 cannot be perfect square.

$\Rightarrow [\sqrt{4k+5}] = [\sqrt{4k+6}]$

By Sandwich theorem

$\Rightarrow [\sqrt{n} + \sqrt{n+1}] = [\sqrt{4n+1}]$

Q. **Let a, b, c be positive real numbers such that $b^2 - 4ac > 0$ and let $\alpha_1 = c$.**

Prove by induction that $\alpha_{n+1} = \frac{a\alpha_n^2}{\left(b^2 - 2a\left(\alpha_1 + \alpha_2 + \ldots + \alpha_n\right)\right.}$ is well - defined and

$\alpha_{n+1} < \frac{\alpha_n}{2}$ for all $n = 1, 2, \ldots$ (Here, 'well - defined' means that the denominator in the expression for α_{n+1} is not zero **[JEE Adv. 2001]**

Sol. We have a, b, c the + ve real number s. t.
$b^2 - 4ac > 0$; $\alpha_1 = c$

$$P(n) : \alpha_{n+1} = \frac{a\alpha_n^2}{b^2 - 2a\left(\alpha_1 + \alpha_2 + \ldots . \alpha_n\right)}$$

is well defined and $\alpha_{n+1} < \frac{\alpha_n}{2}, \forall n = 1, 2, \ldots$

For $n = 1, \alpha_2 = \frac{a\alpha_1^2}{b^2 - 2a\alpha_1} = \frac{ac^2}{b^2 - 2ac}$

Now, $b^2 - 4ac > 0 \Rightarrow b^2 - 2ac > 2ac > 0$

$\therefore \alpha_3$ is well defined (as denomination is not zero)

$$\text{Also} \begin{bmatrix} \because & b^2 - 2ac > 2ac \\ \Rightarrow & \frac{1}{b^2 - 2ac} < \frac{1}{2ac} \end{bmatrix} \Rightarrow \frac{\alpha_2}{c} < \frac{1}{2} \Rightarrow \frac{\alpha_2}{\alpha_1} < \frac{1}{2}$$

$\therefore$ P(n) is true for n = 1

Let the statement be true for $1 \le n \le K$

$$\text{i.e., } \alpha_{k+1} = \frac{a\alpha_k^2}{b^2 - 2a\left(\alpha_1 + \alpha_2 + \ldots \alpha_k\right)} \text{ is well defined}$$

and $\alpha_{k+1} < \frac{\alpha_k}{2}$

Now, we will prove that P(k + 1) is also true

i.e., $\alpha_{k+2} = \dfrac{a\alpha_{k+1}^2}{b^2 - 2a\left(\alpha_1 + \alpha_2 + \ldots + \alpha_k + \alpha_{k+1}\right)}$ is

well defined and $\alpha_{k+2} < \dfrac{\alpha_{k+1}}{2}$.

We have

$$\alpha_1 = c, \alpha_2 < \frac{c}{2}, \alpha_3 < \frac{\alpha_2}{2} < \frac{c}{2^2}, \alpha_4 < \frac{\alpha_3}{2} < \frac{c}{2^3}$$

Now, $\left(\alpha_1 + \alpha_2 + \ldots \alpha_k + \alpha_{k+1} < c + \dfrac{c}{2} + \dfrac{c}{2^k} + \ldots + \dfrac{c}{2^k}\right)$

$$= \frac{c\left(1 - \dfrac{1}{2^{k+1}}\right)}{1 - 1/2} = 2c\left(1 - \frac{1}{2^{k+1}}\right) < 2c$$

$\therefore \quad \alpha_1 + \alpha_2 + \ldots + \alpha_{k+1} < 2c$

$\Rightarrow -2a\left(\alpha_1 + \alpha_2 + \ldots\ldots + \alpha_{k+1}\right) > -4ac$

$\Rightarrow b^2 - 2a\left(\alpha_1 + \alpha_2 + \ldots\ldots + \alpha_{k+1}\right) > b^2 - 4ac > 0$

$\therefore \quad \alpha_{k+2}$ is well defined. Again we have

$\alpha_{k+1} < \dfrac{\alpha_k}{2} \Rightarrow 2\alpha_{k+1} < \alpha_k$

$\Rightarrow 4\alpha_{k+1}^2 < \alpha_k^2$ [As by def. α_{k+1}, α_k are + ve]

$\Rightarrow 4\alpha_{k+1} < \dfrac{\alpha_k^2}{\alpha_{k+1}}$

$\Rightarrow 4\alpha_{k+1} < \dfrac{b^2 - 2a\left(\alpha_1 + \alpha_2 + \ldots + \alpha_k\right)}{a}$

$\Rightarrow 4a\alpha_{k+1} < b^2 - 2a\left(\alpha_1 + \alpha_2 + \ldots + \alpha_k\right)$

$\Rightarrow 2a\alpha_{k+1} < b^2 - 2a\left(\alpha_1 + \alpha_2 + \ldots + \alpha_k + \alpha_{k+1}\right)$

$\Rightarrow \dfrac{a\alpha_{k+1}}{b^2 - 2a\left(\alpha_1 + \alpha_2 + \ldots + \alpha_{k+1}\right)} < \dfrac{1}{2}$

$\Rightarrow \dfrac{a\alpha_{k+1}^2}{b^2 - 2a\left(\alpha_1 + \alpha_2 + \ldots + \alpha_{k+1}\right)} < \dfrac{\alpha_{k+1}}{2}$

$\Rightarrow \alpha_{k+2} < \frac{\alpha_{k+1}}{2}$

$\therefore$ P(k + 1) is also true

Thus by the Principle of Mathematical Induction the statement P(n) is true $\forall\ n \in N$.

Q. **Use mathematical induction to show that $(25)^{n+1} - 24n + 5735$ is divisible by $(24)^2$ for all n = 1, 2.....** **[JEE Adv. 2002]**

Sol. Let P(n) : $(25)^{n+1} - 24n + 5735$

For n = 1

P(1) : 625 -24 + 5735 = 6336 = $(24)^2 \times (11)$,

Which is divisible by 24^2. Hence P(1) is true

Let P(k) be true, where $k \geq 1$

$\Rightarrow (25)^{k+1} - 24k + 5735 = (24)^2\lambda$ where $\lambda \in N$

For n = k + 1 , P(k + 1) : $(25)^{k+2} - 24(k + 1) + 5735$

$= 25[(25)^{k+1} - 24k + 5735] + 25.24.\ k - (25)(5735) + 5735 - 24(k + 1)$

$= 25(24)^2\lambda + (24)^2\ k - 5735 \times 24 - 24$

$= 25(24)^2\ \lambda + (24)^2k - (24)(5736)$

$= 25(24)^2\lambda + (24)^2\ k - (24)^2(239)$

$= (24)^2[25\lambda + k - 239]$ which is divisible by $(24)^2$

Hence, by the method of mathematical induction result is true $\forall\ n \in N$.

Q. **If $a_n = \sqrt{7 + \sqrt{7 + \sqrt{7 + ...}}}$ having n radical signs then by methods of mathematical induction which is true.** **[JEE Adv. 2002]**

(A) $a_n > 7\ \forall\ n \geq 1$ **(B) $a_n < 7\ \forall\ n \geq 1$**

(C) $a_n < 4\ \forall\ n \geq 1$ **(D) $a_n < 3\ \forall\ n \geq 1$**

Sol. **(B)**

$a_1 = \sqrt{7} < 7$. Let $a_m < 7$

Then $a_{m+1} = \sqrt{7 + a_m} \Rightarrow a_{m+1}^2 = 7 + a_m < 7 + 7 < 14$

$\Rightarrow a_{m+1} < \sqrt{14} < 7$;

So by the principle of mathematical induction $a_n < 7\ \forall\ n$

Q. **If $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then which one of the following holds for all $n \geq 1$, by the principle of mathematical induction** **[JEE Adv. 2005]**

(A) $A^n = nA - (n-1)I$ **(B)** $A^n = 2^{n-1}A - (n-1)I$

(C) $A^n = nA + (n-1)I$ **(D)** $A^n = 2^{n-1}A + (n-1)I$

Sol. **(A)**

We observe that

$A^2 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, A^3 = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$ and we can prove by induction that $A^n = \begin{bmatrix} 1 & 0 \\ n & 1 \end{bmatrix}$

Now $nA - (n-1)I = \begin{bmatrix} n & 0 \\ n & n \end{bmatrix} - \begin{bmatrix} n-1 & 0 \\ 0 & n-1 \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 \\ n & 1 \end{bmatrix} = A^n$

$\therefore nA - (n-1)I = A^n$